[INCH-POUND]
GGG-C-744B
March 27, 1996
SUPERSEDING
GGG-C-744A
May 15, 1965

FEDERAL SPECIFICATION

CUTTER AND FLARING TOOL KIT, TUBE, HAND;

CUTTER, FLARING AND BENDER TOOL KIT, TUBE HAND

The General Services Administration has authorized the use of this Federal Specification by all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers kits of hand-operated tools used in cutting, flaring, and bending soft steel, copper, and aluminum tubing.

1.1.1 Federal specification coverage. Federal specifications do not cover all varieties of the commodity indicated by the title of the specification, or which are commercially available, but are intended to cover only those generally used by the Federal Government.

1.2 Classification.

1.2.1 Types. Tool kits covered by this specification shall be of the following types (see 6.2):

Type I - Cutter and flaring tool kit, tube hand, flaring capacity 3/16 inch - 5/8 inch O.D. tubing

Type II - Cutter, flaring, and bender tool kit, tube hand, flaring capacity 1/8 inch - 3/4 inch O.D. tubing

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

GGG-B-191	-	Bender, Tube, Hand, (Portable)
GGG-C-771	-	Cutters, Pipe; and Cutters, Tube
GGG-F-451	-	Flaring Tool, Tube, Hand
PPP-P-40	-	Preservation and Packing of Hand Tools and Tool Accessories for Power Driven, Metal and Woodworking Machinery

Beneficial comments, recommendations, additions, deletions, clarifications, etc., and any other data which may improve this document should be sent to: General Services Administration, Federal Supply Service, Tools and Appliances Commodity Center (6FETE-CO), Washington, DC 20406.

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Federal Standard:

FED-STD-151 - Metal, Test Methods

(Activities outside the Federal Government may obtain copies of Federal specifications, standards, and Commercial Item Descriptions as outlined under General Information in the Index of Federal Specifications, Standards and Commercial Item Descriptions. The Index, which includes cumulative bimonthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

(Single copies of this specification, and other Federal specifications and Commercial Item Descriptions required by activities outside the Federal Government for bidding purposes are available without charge from General Services Administration Business Service Centers in Boston, MA; New York, NY; Philadelphia, PA; Washington, DC; Atlanta, GA; Chicago, IL; Kansas City, MO; Fort Worth, TX; Houston, TX; Denver, CO; San Francisco, CA; Los Angeles, CA; and Seattle, WA.)

(Federal Government activities may obtain copies of Federal standardization documents and the Index of Federal Specifications, Standards and Commercial Item Descriptions from established distribution points in their agencies.)

2.2 Non-Government publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American National Standards Institute (ANSI)/American Society of Quality Control (ASQC):

ANSI/ASQC Z1.4 - Sampling Procedures and Tables for Inspection by Attributes

(Application for copies should be addressed to the American Society for Quality Control, P.O. Box 3005, 611 E. Wisconsin Avenue, Milwaukee, WI 53201-4606.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

3. REQUIREMENTS

3.1 Illustrations. Illustrations herein are descriptive and not restrictive and are not intended to preclude the purchase of tool kits otherwise conforming to this specification.

3.2 Design. The cutter, flaring, and bending tool kit components shall be in accordance with referenced Federal specifications or this specification.

3.3 Type I cutter and flaring tool kit. Type I cutter and flaring tool kit shall have the following components.

3.3.1 Tube cutter. The tube cutter shall be the type II, class 1, tube cutter of GGG-C-771, having a cutting capacity of 1/8 inch to 1-1/8 inches outside the diameter of tubing.

3.3.2 Flaring tool. The flaring tool shall be the type I, class 2, style A, 90° flaring tool, tool no. F of GGG-F-451.

3.3.3 Tool case. The tool case shall be of a convenient size and shape with adequate means for holding all components of the kit. The case shall be of sturdy construction to withstand the test in 4.5.2. The case shall be made of metal or other suitable material, with a hinged lid and a positive catch. A protective coat of paint shall be applied on both the inside and outside of the metal cases.

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3.4 Type II cutter, flaring, and bending tool kit. Type II cutter, flaring, and bending tool kit shall have the following components.

3.4.1 Tube cutter. The cutter shall be in accordance with 3.3.1.

3.4.2 Flaring tool. The tool shall be the type I, class 2, style A, 90° flaring tool, tool no. J of GGG-F-451.

3.4.3 Tube benders. The benders shall be type IV of GGG-B-191 for 1/4 inch, 3/8 inch, and 1/2 inch sizes.

3.4.4 Ratchet wrench.

3.4.4.1 Material. The handle shall be a steel forging and the ratchet socket shall be of steel. The chemical composition of the steels and the heat treatment employed shall be such as to produce a ratchet socket hardness of 47 to 52 of the Rockwell C scale and a handle hardness of 41 to 45 on the Rockwell C scale.

3.4.4.2 Construction. The ratchet wrench shall consist of a handle, ratchet mechanism, and a ratchet socket. The ratchet socket shall have a 1/4 inch or 3/8 inch square opening and the handle shall have 3/16 inch square, 1/4 inch square and 1/2 inch hexagon openings. The handle shall securely house the ratchet mechanism and ratchet socket, and also allow the ratchet socket to rotate without binding. All surfaces which bear or mate with the ratchet socket shall have a machine finish. The reversing of the direction of the ratcheting operation shall be accomplished either by a shifting device or by the operator turning the wrench over. The ratchet socket shall have a minimum of 16 ratchet teeth. The ratchet wrench shall have a minimum plating thickness of 0.0002 inch of nickel followed by a minimum thickness of 0.000007 inch of chromium. The wrench shall conform to Table I and be similar to Figure 1.

TABLE I. Ratchet wrench

Overall length	Wrench Openings (Distance across flats)						Test load	
	A		B		C		Ratchet Socket A	1/2 inch hexagon C
Minimum (Inches)	Minimum (Inch)	Maximum (Inch)	Minimum (Inch)	Maximum (Inch)	Minimum (Inch)	Maximum (Inch)	Minimum (Inch-pounds)	Maximum (Inch-pounds)
5-3/4	0.251	0.258	0.189	0.196	0.503	0.511	150	500
5-3/4	0.377	0.385	0.189	0.196	0.503	0.511	150	500

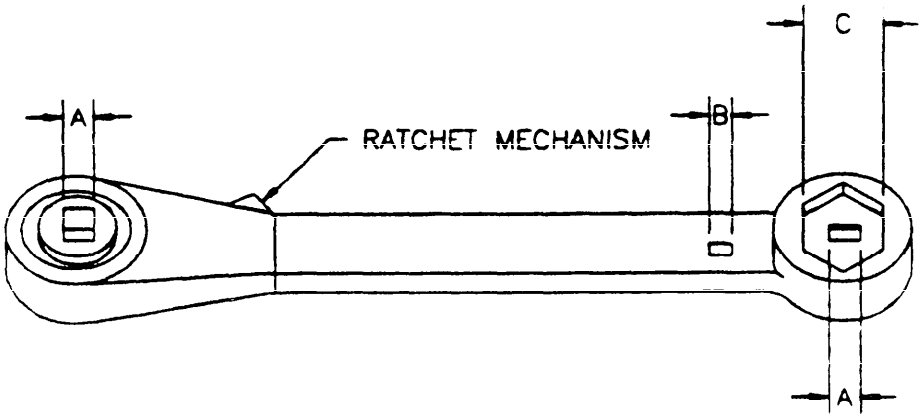


Figure 1. Ratchet wrench

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3.5 Marking. Each item shall be marked with the manufacturer's name or identifying symbol and the state or country of manufacture. The marking shall be engraved, etched, or stamped in such a manner that it will be permanent to the extent that it will remain clear and legible throughout the life of the item.

3.6 Tool case. The tool case shall be in accordance with 3.3.3.

3.6.1 Instruction sheet. An instruction sheet shall be cemented to the inside cover of each tool case.

3.7 Repair parts. The successful bidders supplying the kits covered by this specification shall furnish repair parts when specified in the contract or order or when requested (see 6.1). All parts having the same manufacturer's part number shall be constructed to definite standards, tolerances, and clearances in order that such parts may be completely interchangeable, replaced, or adjusted without requiring modification.

3.8 Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within the tolerances specified and all other requirements of this document are met. If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch-pound units, a request should be made to the specification preparing activity for change to this document.

3.9 Workmanship: Details of workmanship shall be in accordance with the best commercial standards and practices. Paints, coatings, platings, and finishes shall be smooth, dry, adherent, continuous, and not stained or discolored. Fasteners shall be firmly secured and show no evidence of deformation, cross threading, or hazardous burrs. Adhesives and lubricants adequate for their intended purpose shall be properly and neatly applied. Adhesives shall be adequately cured. Wires and cables shall be neatly dressed and shall not be frayed or in contact with sharp edges. Wire and cable insulation shall show no evidence of burns, abrading, or pinch marks. There shall be no interference, binding, or galling of parts. External and bearing surfaces shall be free of tool and gouge marks, nicks, or other surface imperfections. The item shall be clean and free of corrosion and debris (e.g., chips, shavings, slivers) or other foreign material. The item shall be free from manufacturing workmanship defects (e.g., loose, missing, binding or misaligned parts, sharp or rough external edges, corners, or surfaces) and material workmanship defects (e.g., pits, rips, fins, burrs, tears, nodules, cracks, blisters) which may adversely impact the item's serviceability, durability, safety, or appearance.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. In accordance with 4.1 above, the supplier is responsible for insuring that components and materials used were manufactured, tested, and inspected in accordance with the requirements of referenced subsidiary specification and standards to the extent specified, or, if none, in accordance with this specification.

4.2 Sampling procedures. Unless otherwise specified, sampling for inspection shall be performed in accordance with ANSI/ASQC Z1.4. Data for sampling shall be as specified in Table II.

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TABLE II. Sampling data

Category	Sample unit	Inspection level	Acceptable quality level (AQL)	AQL expressed in terms of	Reference
Visual examination of tool case or ratchet wrench	1 each	II	4.0	Defects per hundred units	4.4.1
Visual examination of assembled kit	1 kit	S-2	1.0	Defects per hundred units	4.4.2
Dimensional examination for ratchet wrench	1 each	S-4	1.5	Defects per hundred units	4.4.4
Testing of ratchet wrench	1 each	S-4	1.0	Percent defective	4.4.1
Testing of case	1 each	S-1	1.0	Percent defective	4.4.2
Preparation for delivery	One container	S-2	4.0	Defects per hundred units	4.5

NOTE: Type II, class 1 tube cutters shall be inspected in accordance with GGG-C-771.
Type I, class 2, style A flaring tools shall be inspected in accordance with GGG-F-451.
Type IV tube benders shall be inspected in accordance with GGG-B-191.
Repair parts, when furnished, shall be visually examined for compliance with applicable specification requirements using sampling data specified in Table II.
No kit shall be accepted unless all components pass the inspection criteria specified herein.

4.3 Examination.

4.3.1 Visual examination of case or ratchet wrench. Each sample unit shall be examined for any nonconformance in design, material, finish, coating, construction, workmanship, and marking.

4.3.2 Visual examination kit. Each sample kit shall be examined for completeness.

4.3.3 Dimensional examination. Each sample unit shall be examined for any nonconformance with dimensional requirements.

4.4 Testing of case or ratchet wrench. Each sample unit shall be tested in accordance with 4.4.1 or 4.4.2, as applicable.

4.4.1 Ratchet wrenches. Each of the sample ratchet wrenches shall be tested for compliance with the test load of Table I, and the hardness requirements of 3.3.4.2. Hardness testing shall be performed in accordance with method 243.1 of FED-STD-151.

4.4.2 Tool case. Each of the sample cases (complete with tools) shall be dropped five times from a height of 4 feet onto a concrete floor or slab. At the conclusion of the drop test there shall be no evidence of permanent damage to the case or contents. Any sample failing to meet this test shall cause rejection of the lot represented by the sample.

4.5 Inspection of preparation for delivery requirements. An inspection shall be made to determine that the preservation, packaging, packing, and marking comply with the requirements of PPP-P-40.

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5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing, and marking. Preservation, packaging, packing, and marking shall be in accordance with PPP-P-40. The level of preservation, packaging, and packing shall be A, B, or Commercial, as specified (see 6.1).

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

6.1 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Type of item required (see 1.2).
c. Whether repair parts are required (see 3.7)
d. Level of preservation, packaging, and packing required (see 5.1).

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